Question	The equation of a transverse wave is given by $y = 10\sin\pi(0.01x - 2t)$ Where x and y are in cm and t is in second. Its frequency is
Type	multiple choice
Option	$10\ \text{sec}^{-1}$
Option	$2\ \text{sec}^{-1}$
Option	$1\ \text{sec}^{-1}$
Option	$0.01\ \text{sec}^{-1}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following equations represents a wave travelling along y - axis
Type	multiple choice
Option	$y = A\sin(kx - \omega t)$
Option	$y = A\sin(ky - \omega t)$
Option	$y = A\sin ky\ \cos\omega t$
Option	$y = A\cos ky\sin\omega t$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two waves are given by $y_1(\omega t - kx)$ and $y_2 a\cos(\omega t - kx)$ The phase difference between the two wave is
Type	multiple choice
Option	$\pi/4$
Option	π
Option	$\pi/8$
Option	$\pi/2$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A wave travelling in positive x direction with A = 0.2 m has a velocity of 360 m/sec. If $\lambda = 60\,m,$ then correct expression for the wave is
Type	multiple choice
Option	$y = 0.2\sin\left[2\pi\left(6t + \frac{x}{60}\right)\right]$
Option	$y = 0.2\sin\left[\left(6t + \frac{x}{60}\right)\right]$
Option	$y = 0.2\sin\left[2\pi\left(6t - \frac{x}{60}\right)\right]$
Option	$y = 0.2\sin\left[\left(6t - \frac{x}{60}\right)\right]$
Answer	3
Solution	
Positive Marks	4

Negative Marks	1
Question	If the wave equation is given by $y = 0.08 \ \sin\frac{2\pi}{\lambda}(200t - x)$, then the velocity of the wave will be (in m/s)
Type	multiple choice
Option	$400\sqrt{2}$
Option	$200\sqrt{2}$
Option	400
Option	200
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The equation of a sound wave is $y = 0.0015\sin(62.8x + 316t)$ The wavelength of this wave is
Type	multiple choice
Option	0.2 unit
Option	0.1 unit
Option	0.3 unit
Option	Cannot be calculated
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Equation of a progressive wave is given by $y = a\sin\left[\frac{t}{2} - \frac{x}{4}\right]$, where t is in second and x is in meters. The distance through which the waves move in 8 sec is (in meter)
Type	multiple choice
Option	8
Option	16
Option	2
Option	4
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	When a wave travels in a medium, the particle displacement is given by the equation $y = a\sin 2\pi(bt - cx)$ where a, b and c are constants. The maximum particle velocity will be twice the wave velocity if
Type	multiple choice
Option	$c = \frac{1}{\pi a}$
Option	$c = \pi a$
Option	$b = ac$
Option	$b = \frac{1}{ac}$
Answer	1
Solution	
Positive Marks	4

Negative Marks	1
Question	The phase difference between two waves represented by $y_1 = 10^{-6}\sin\left[100t + (x/50) + 0.5\right]m$ $y_2 = 10^{-6}\cos\left[100t + (x/50)\right]m$ (where x is expressed in meters and t is expressed in seconds) is approximately
Type	multiple choice
Option	1.5 rad
Option	1.07 rad
Option	2.07 rad
Option	0.5 rad
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A string wave $y = A_0 \sin(\omega t - kx)$ is reflected from a rigid wall with 64% of its amplitude. The equation of the reflected wave is
Type	multiple choice
Option	$y = \frac{64}{100} A_0 \sin(\omega t + kx)$
Option	$y = -\frac{64}{100} A_0 \sin(\omega t + kx)$
Option	$y = \frac{64}{100} A_0 \sin(\omega t - kx)$
Option	$y = -\frac{64}{100} A_0 \cos(\omega t - kx)$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A string is producing transverse vibration whose equation is $y = 0.021 \sin(x + 30t)$. where x and y are in meters and t is in seconds. If the linear density of the string is $1.3 \times 10^{-4} kg/m$, then the tension in the string in N will be
Type	multiple choice
Option	10
Option	0.5
Option	1
Option	0.117
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The speed of a wave on a string is 150 m/s when the tension is 120 N. The percentage increase in the tension in order to raise the wave speed by 20% is
Type	multiple choice
Option	44%
Option	40%
Option	20%
Option	10%
Answer	1
Solution	

Positive Marks	4
Negative Marks	1
Question	Transvers waves of same frequency are generated in two steel wires A and B. The diameter of A is twice of B and the tension in A is half that in B. The ratio of velocities of wave in A and B is
Type	multiple choice
Option	$1:3\sqrt{2}$
Option	$1:2\sqrt{2}$
Option	$1:2$
Option	$\sqrt{2}:1$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The tension in a piano wire is 1ON. What should be the tension in the wire to produce a note of double the frequency
Type	multiple choice
Option	5 N
Option	20 N
Option	40 N
Option	80 N
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The speed of sound in a gas of density ρ at a pressure P is proportional to
Type	multiple choice
Option	$\left(\frac{p}{\rho}\right)^2$
Option	$\left(\frac{p}{\rho}\right)^{3/2}$
Option	$\sqrt{\frac{\rho}{p}}$
Option	$\sqrt{\frac{P}{\rho}}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The ratio of the speed of sound in nitrogen gas to that in helium gas, at 300 K is
Type	multiple choice
Option	$\sqrt{\frac{2}{7}}$
Option	$\sqrt{\frac{1}{7}}$
Option	$\frac{\sqrt{3}}{5}$

Option	$\frac{\sqrt{6}}{5}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The speed of sound in oxygen (O_2) at a certain temperature is $460\,ms^{-1}$.The speed of sound in helium (He) at the same temperature will be (assume both gases to be ideal)
Type	multiple choice
Option	$500\,ms^{-1}$
Option	$650\,ms^{-1}$
Option	$330\,ms^{-1}$
Option	$1420\,ms^{-1}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The velocity of sound is v_s in air. If the density of air is increased to 4 times, then the new velocity of sound will be
Type	multiple choice
Option	$\frac{v_s}{2}$
Option	$\frac{v_s}{12}$
Option	$12v_s$
Option	$\frac{3}{2}v_s^2$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	When the temperature of an ideal gas is increased by 600 K, the velocity of sound in the gas becomes $\sqrt{3}$ times the initial velocity in it. The initial temperature of the gas is
Type	multiple choice
Option	-73^0C
Option	27^0C
Option	127^0C
Option	327^0C
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two monoatomic ideal gases 1 and 2 of molecular masses m_1 and m_2 respectively are enclosed in separate containers kept at the same temperature. The ratio of the speed of sound in gas 1 to that in gas 2 is given by

Type	multiple choice
Option	$\sqrt{\frac{m_1}{m_2}}$
Option	$\sqrt{\frac{m_2}{m_1}}$
Option	$\frac{m_1}{m_2}$
Option	$\frac{m_2}{m_1}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The amplitude of a wave represented by displacement equation $y=\frac{1}{\sqrt{a}}\sin\omega t\pm\frac{1}{\sqrt{b}}\cos\omega t$ will be
Type	multiple choice
Option	$\frac{a+b}{ab}$
Option	$\frac{\sqrt{a}+\sqrt{b}}{ab}$
Option	$\frac{\sqrt{a}\pm\sqrt{b}}{ab}$
Option	$\sqrt{\frac{a+b}{ab}}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Two waves having equations $x_1=a\sin(\omega t+\phi_1), x_2=a\sin(\omega t+\phi_2)$ If in the resultant wave the frequency and amplitude remain equal to those of superimposing waves. Then phase difference between them is
Type	multiple choice
Option	$\pi/6$
Option	$2\pi/3$
Option	$\pi/4$
Option	$\pi/3$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two waves coming from two coherent sources, having different intensities interfere. The ratio of maximum intensity to the minimum intensity is 25. The intensities of the sources are in the ratio
Type	multiple choice
Option	$25:1$
Option	$25:16$

Option	$9:4$
Option	$5:1$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Two waves having the intensities in the ratio of 9 : 1 produce interference. The ratio of maximum to the minimum intensity, is equal to
Type	multiple choice
Option	$2:1$
Option	$4:1$
Option	$9:1$
Option	$10:8$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Equation of motion in the same direction are given by $y_1 = 2a\sin(\omega t - kx)$ and $y_2 = 2a\sin(\omega t - kx - \theta)$ The amplitude of the medium particle will be
Type	multiple choice
Option	$2a\cos\theta$
Option	$\sqrt{2}a\cos\theta$
Option	$4a\cos\theta/2$
Option	$\sqrt{2}a\cos\theta/2$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The superposing waves are represented by the following equations: $y_1 = 5\sin 2\pi(10t - 0.1x)$ $y_2 = 10\sin 2\pi(10t - 0.1x)$ Ratio of intensities $\frac{I_{max}}{I_{min}}$ will be
Type	multiple choice
Option	1
Option	9
Option	4
Option	16
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two identical sinusoidal waves each of amplitude 5 mm with a phase difference of $\pi/2$ are traveling in the same direction in a string. The amplitude of the resultant wave (in mm) is
Type	multiple choice
Option	Zero
Option	$5\sqrt{2}$
Option	$5/\sqrt{2}$
Option	2.5
Answer	2

Solution	
Positive Marks	4
Negative Marks	1
Question	Two periodic waves of intensities I_1 and I_2 pass through a region at the same time in the same direction. The sum of the maximum and minimum intensities is
Type	multiple choice
Option	$\left(\sqrt{I_1}-\sqrt{I_2}\right)^2$
Option	$2\left(I_1+I_2\right)$
Option	I_1+I_2
Option	$\left(\sqrt{I_1}+\sqrt{I_2}\right)^2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The displacement of the interfering sound waves are $y_1=4\sin\omega t$ and $y_2=3\sin\left(\omega t+\frac{\pi}{2}\right)$. What is the amplitude of the resultant wave
Type	multiple choice
Option	5
Option	7
Option	1
Option	0
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Two sound waves travel in the same direction in a medium. The amplitude of each wave is A and the phase difference between the two waves is 120^0. The resultant amplitude will be
Type	multiple choice
Option	$\sqrt{2}A$
Option	$2A$
Option	$3A$
Option	A
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The phase difference between the two particles situated on both the sides of a node is
Type	multiple choice
Option	0^0
Option	90^0
Option	180^0
Option	360^0
Answer	3
Solution	
Positive Marks	4

Negative Marks	1
Question	Consider the three waves z_1, z_2 and z_3 as $z_1 = A\sin(kx - \omega t)$, $z_2 = A\sin(kx + \omega t)$ and $z_3 = A\sin(ky - \omega t)$. Which of the following represents a standing wave
Type	multiple choice
Option	$z_1 + z_2$
Option	$z_2 + z_3$
Option	$z_3 + z_1$
Option	$z_1 + z_2 + z_3$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	When a stationary wave is formed then its frequency is
Type	multiple choice
Option	Same as that of the individual waves
Option	Twice that of the individual waves
Option	Half that of the individual waves
Option	None of the above
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The equation of stationary wave along a stretched is given by $y = 5\sin\frac{\pi x}{3}\cos 40\pi t$, where x and y are in cm and t in second. The separation between two adjacent nodes is
Type	multiple choice
Option	1.5 cm
Option	3 cm
Option	6 cm
Option	4 cm
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A wave of wavelength 2m is reflected from a surface. If a node is formed at 3m from the surface, then at what distance from the surface another node will be formed
Type	multiple choice
Option	3.25 m
Option	3.5 m
Option	3.75 m
Option	4 m
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The fundamental frequency of a closed organ pipe of length 20 cm is equal to the second overtone of an organ pipe open at both the ends. The length of organ pipe open at both the ends is

Type	multiple choice
Option	100 cm
Option	120 cm
Option	140 cm
Option	80 cm
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A closed pipe and an open pipe have their first overtones identical in frequency. Their lengths are in the ratio
Type	multiple choice
Option	$1:2$
Option	$2:3$
Option	$3:4$
Option	$4:5$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The length of an open pipe is twice the length of another closed organ pipe. The fundamental frequency of the open pipe is 100 Hz. The frequency of the third harmonic of the closed pipe is
Type	multiple choice
Option	$100\ Hz$
Option	$200\ Hz$
Option	$300\ Hz$
Option	$150\ Hz$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If the length of a closed organ pipe is 1 m and velocity of sound is 330 m/s, then the frequency for the second note is
Type	multiple choice
Option	$400\ Hz$
Option	$247.5\ Hz$
Option	$350.5\ Hz$
Option	$750.5\ Hz$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If the length of a open organ pipe is 1.5 m and velocity of sound is 330 m/s, then the frequency for the fourth overtone is
Type	multiple choice
Option	$220\ Hz$
Option	$165\ Hz$
Option	$110\ Hz$
Option	$550\ Hz$
Answer	4

Solution	
Positive Marks	4
Negative Marks	1
Question	The fundamental frequency of closed pipe is 50 Hz then frequency of 2nd overtone is
Type	multiple choice
Option	$100\ Hz$
Option	$50\ Hz$
Option	$250\ Hz$
Option	$150\ Hz$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Two closed pipes produce 10 beats per second when emitting their fundamental nodes. If their lengths are in ratio of 25: 26. Then their fundamental frequency in Hz, are
Type	multiple choice
Option	$270, 280$
Option	$260, 270$
Option	$260, 250$
Option	$260, 280$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Two closed organ pipe of length 100 cm and 101 cm produce 16 beats in 20 sec. When each pipe is sounded in its fundamental mode calculate the velocity of sound
Type	multiple choice
Option	$303\ ms^{-1}$
Option	$332\ ms^{-1}$
Option	$323.2\ ms^{-1}$
Option	$300\ ms^{-1}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A standing wave is represented by $Y = A\sin(100t)\cos(0.01x)$ where Y and A are in millimeter, t is in seconds and x is in metre. The velocity of wave is
Type	multiple choice
Option	$10^4\ m/s$
Option	$1 m/s$
Option	$10^{-4}\ m/s$
Option	Not derivable from above data
Answer	1
Solution	
Positive Marks	4

Negative Marks	1
Question	Two uniform strings A and B made of steel are fixed from both ends and are made to vibrate under the same tension. If the first overtone of A is equal to the second overtone of B and if the radius of A is twice that of B, the ratio of the lengths of the string is
Type	multiple choice
Option	$1:2$
Option	$3:2$
Option	$3:4$
Option	$1:3$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1